

COMICCRAFT

Phase 2 – Requirement Analysis

Project: ComicCraft – AI Comic Story Creator using Gemini Models

1. Purpose of Requirement Analysis

The Requirement Analysis Phase converts the ideas identified during brainstorming into clear requirements for the ComicCraft application. The project documentation describes ComicCraft as a web-based application that accepts user information and generates a panel-by-panel storyline with corresponding illustrations, followed by comic preview and PDF export.

2. Core Project Requirement

The system should provide an end-to-end comic generation workflow. It should collect the user's creative inputs, generate structured comic content, generate illustrations, display the result, and support export of the comic as a PDF.

3. Main User Inputs

- Story Prompt – the main idea for the comic.
- Character Name – the hero or main character.
- Setting – the location or environment.
- Story Tone – the mood of the story.
- Art Style – the preferred visual style.

4. Main Expected Outputs

- Structured comic panel outline.
- Comic narration and character dialogue.
- Comic-style illustrations for panels.
- Panel-by-panel comic preview.
- Exportable PDF containing images and narration.

Functional Requirements

5. Functional Requirements

Functional requirements describe what the ComicCraft system must do during the comic generation process.

ID	Requirement
FR-01	The system shall provide a web interface for entering comic information.
FR-02	The system shall accept story prompt, character name, setting, tone, and art style.
FR-03	The system shall process the submitted user information through the backend.
FR-04	The system shall generate a structured panel-by-panel comic outline.
FR-05	The system shall generate narration and character dialogue.
FR-06	The system shall generate comic-style illustrations from image prompts.
FR-07	The system shall organize generated images and story content into a comic layout.
FR-08	The system shall display the generated comic through a preview page.
FR-09	The system shall compile the comic into a PDF file.
FR-10	The system shall provide an export/download flow for the generated comic.

6. AI Processing Requirements

- Gemini Flash is identified for structured comic panel outline generation.
- Gemini Pro is identified for detailed narration and character dialogue.
- Stable Diffusion is identified for comic-style illustration generation.
- The backend must coordinate the generated text and images before sending them to the frontend.

Non-Functional & Interface Requirements

7. Non-Functional Requirements

- **Usability:** The interface should be simple and user-friendly so that non-artists and casual users can create comics.
- **Accessibility of workflow:** The creation process should move clearly from user input to comic preview and export.
- **Responsiveness:** The frontend should provide an organized layout and readable comic panels.
- **Maintainability:** The application should keep frontend, backend, AI integration, image generation, and export responsibilities organized into separate components.
- **Development environment:** The application should run locally using the FastAPI/Uvicorn development setup described in the project documentation.

8. Frontend Requirements

- index.html should collect the required user inputs.
- comic_preview.html should display generated panels, images, scene descriptions, captions, and narration.
- export_success.html should provide confirmation after comic export.
- Jinja2 templates should dynamically receive generated backend data.

9. Backend Requirements

- FastAPI should handle routing and user input.
- Form data should be captured using FastAPI form handling.
- Pydantic PromptRequest should support structured API input where applicable.
- Backend routes should connect the comic generation workflow with the AI functions.

10. API/Route Requirements from the Project Documentation

- / – loads the homepage.
- /generate – handles comic generation from form input.
- /generate-comic/json – supports JSON-based comic generation.
- /test-image – supports direct image-generation testing.
- /export-success – displays export confirmation.

Technology, Data & Processing Requirements

11. Technology Requirements

Area	Required Technology
Programming	Python
Web Framework	FastAPI
Server	Uvicorn ASGI Server
Frontend	HTML, CSS, Jinja2
Text AI	Google Gemini models
Image AI	Stable Diffusion / Hugging Face Diffusers
Image Support	Pillow
PDF Export	FPDF
Version Control	Git
Environment	Python virtual environment and pip

12. Data Processing Requirements

- Receive and validate user-provided story information.
- Create structured panel outlines from the story prompt.
- Expand outlines into narration and dialogue.
- Generate image prompts and corresponding illustrations.
- Associate each generated image with its panel story.
- Build the final comic layout.
- Pass the completed layout to the PDF export process.

13. File and Environment Requirements

The project documentation specifies a virtual environment, dependency installation through pip, and environment variables for sensitive API credentials. The documented examples include a Gemini API key and a Hugging Face API token. These values should be kept outside publicly shared source code.

14. Development Environment Requirement

The application is expected to be runnable from the project root using Uvicorn with the FastAPI application entry point. The project documentation identifies the local application URL and the FastAPI interactive documentation URL for testing.

Requirement Summary & Acceptance Criteria

15. Requirement Summary

The requirement analysis establishes the complete functional flow needed for ComicCraft: collect user preferences, generate structured story content, generate illustrations, display the comic, and export the final result. The requirements also establish the technology stack and major application components.

16. Basic Acceptance Criteria

No.	The system should be able to...
1	Accept the five main comic inputs from the user.
2	Process the request through the FastAPI backend.
3	Generate a structured comic outline.
4	Generate narration and dialogue.
5	Generate corresponding comic illustrations.
6	Display the generated content panel-by-panel.
7	Organize images and story content into a comic layout.
8	Create a PDF containing the comic content.
9	Provide an export/download confirmation flow.

17. Phase 2 Outcome

At the end of Requirement Analysis, ComicCraft has a defined set of user inputs, functional requirements, non-functional requirements, technology requirements, frontend/backend requirements, AI processing requirements, and acceptance criteria. These requirements provide the foundation for Phase 3 – Project Design.

18. Next Phase

Phase 3 – Project Design will use these requirements to define the system architecture, application workflow, component relationships, UI structure, and project folder design.

Source basis: ComicCraft – AI Comic Story Creator using Gemini Models project documentation.